# THE ROLE OF ESA IN THE EVOLUTION OF SPACE LAW

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The European Space Agency (ESA) is a regional international intergovernmental organization whose mandate is to conduct space research and development for exclusively peaceful purposes.[1] The Agency has currently thirteen European Member States,[2] plus a special cooperative relationship with Canada providing for Canada's participation in several ESA programmes.[3] In 1995 Finland, an Associate State since 1987, will become the fourteenth ESA Member State (the Accession Agreement was signed on 22 March 1994). ESA is a unique organization as it is the only international organization in the world dedicated to the research and development of space activities (since the dissolution of INTERCOSMOS). ESA has legal personality[4] and is therefore subject to international law, especially its space law provisions. ESA accepted by declaration the Rescue Agreement,[5] the Registration Convention[6] and the Liability Convention,[7] and has rights and obligations under these treaties

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1 Art. II of the ESA Convention, *in* 2 SPACE LAW - BASIC LEGAL DOCUMENTS, C. I. (K.-H. Böckstiegel & M. Benkö eds. Nijhoff 1992).

2 Austria, Belgium, Denmark, France, Germany, Ireland, Italy, Netherlands, Norway, Spain, Sweden, Switzerland, United Kingdom.

3 *See* Cooperation Agreement between the Government of Canada and the European Space Agency, signed 31 May 1989, Doc. ESA/LEG 117. See on the relations between ESA and Canada: M. Bourély, *Le Canada et l'Agence Spatiale Européenne*, 4 ANN. AIR & SPACE L. 397 (1979); M. Bourély, *La participation du Canada aux programmes de l'Agence Spatiale Européenne*, 5 ANN. AIR & SPACE L. 363 (1980); G. Lafferranderie, *L'affermissement des relations entre le Canada et l'Agence Spatiale Européenne*, 14 ANN. AIR & SPACE L. 369. (1989); G. Lafferranderie, *The Enlargement of the European Space Agency - Legal Issues*, 15 J. SPACE L. 119 (1987).

4 Art. XV of the ESA Convention, *in* SPACE LAW - BASIC LEGAL DOCUMENTS, *supra* note 1.

5 Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space, 22 April 1968 (hereinafter Rescue Agreement), TIAS 6599, 19 UST 7570.

6 Convention on Registration of Objects Launched into Outer Space, 14 January 1975 (hereinafter Registration Convention), TIAS 8480, 28 UST 695.

7 Convention on International Liability for Damage Caused by Space Objects, 29 March 1972 (hereinafter Liability Convention), TIAS 7762, 24 UST 2389.

independently from the Member States. ESA was associated, through its internal policy-making process, with the European contributions for adopting the United Nations Principles on remote sensing and nuclear power sources.

It is the purpose of this article to describe what kind of legal problems are encountered by ESA when it carries out its activities and what role ESA plays in the further development of space law. Two features of ESA play an important role. ESA is an international cooperative venture striving to honor the principles of the Outer Space Treaty,[8] which provides for international cooperation in the conduct of space activities.[9] However, in the same Article, it is provided that any practical problems should be solved by the States Party to the Outer Space Treaty. This situation raises questions with regard to competing national jurisdictions (of the Member States), liability and responsibility allocations.

In addition, the activities of ESA unavoidably lead to the creation of additional rules of space law or to the practical implementation of the general principles of space law. Further, the creation of new space activities resulting from its R&D efforts inevitably gives ESA a special character, as law follows technical developments and ESA is by definition in the forefront of developments and applications. [10]

Finally, ESA is also involved in programmes of application, and the maturity of these activities reflects the need for more practical down-to-earth rules and solutions whereby space law principles acquire a more empowering role.

That ESA has a specific role in the evolution of space law will be clear from the above. For the future however, given the number of important projects (most of them involving international cooperation) to be implemented, such as the international Space Station, it will be of interest to see which direction ESA will take as an international organization and what influences will be exerted upon space law as a result.

## ESA AND SPACE LAW

ESA succeeded ESRO (European Space Research Organisation) and ELDO (European Launcher Development Organisation) and started to carry out its activities, de facto, in 1975.[11] The ESA Convention came into force

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8 Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and other Celestial Bodies, 27 January 1967 (hereinafter Outer Space Treaty), TIAS 6347, 18 UST 2410.

9 *See* article XIII of the Outer Space Treaty which provides that: "The provisions of this Treaty shall apply to the activities of States Parties to the Treaty in the exploration and use of outer space, including the moon and other celestial bodies, whether such activities are carried on by a single State party to the Treaty or jointly with other States, including cases where they are carried on within the framework of international inter-governmental organizations..."

10 For a similar argument, *see* M. Bourély, *L'Agence Spatiale et le Droit de l'Espace,* ESA Bull.. 39 (No. 13, 13 May 1978).

11 After its approval by the European Space Conference (ESC) on 15 April 1975, the ESA Convention was opened for signature at the Plenipotentiary Conference on 30 May 1975. This concluded the discussions of the ESC which

in 1980. ESA has a twofold structure, part of its activities being mandatory, mainly the science programme, and funded in accordance with the Member States GNP percentages, and part of the programmes (application oriented) being optional where the Member States have the option not to participate. Currently, ESA is executing optional programmes in the following areas: Earth Observation, Telecommunications, (Manned) Space Transportation and Micro-gravity. The optional programmes[12] are the most interesting from the point of view of space law as they raise all kinds of questions regarding the implementation of the general principles of space law. The Council has entrusted the International Relations Advisory Group, renamed Committee and today, the International Relations Committee (IRC), as an advisory body for the formulation of ESA's position, and coordination between the Member States, vis-a-vis the UNCOPUOS. The main role of the IRC is to advise Council on external relations policy, conclusion of agreements with non-Member States, *etc.* IRC also functions as a forum for exchanging views on UN developments.

### *Standard Article on Declaration of Acceptance by International Organizations*

ESA's role in the development of space law goes back to the formulation of the basic principles of space law. The first example of such formulation may be found in the work of the European Space Conference Working Group in charge of the United Nations. This group was at the origin of the drafting and adoption of article VI of the Outer Space Treaty, which was later implemented in the other space treaties, and which provides for the declaration of acceptance of these treaties by international organizations.[13]

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started on 20 December 1972, where the Ministers adopted the concept of a European Space Agency which should continue the activities of ESRO and ELDO. ESRO conducted its activities under the name of ESA and used when possible the provisions of the ESA Convention. *See* M. Bourély, *Les traits saillants de la Convention portant création d'une Agence Spatiale Européenne,* ESA Bull. (1 June 1975), reprinted in EUROPEAN CENTRE FOR SPACE LAW (ECSL), SPACE LAW AND POLICY SUMMER COURSE - BASIC MATERIALS (Nijhoff 1993); M. Bourély, *La naissance de l'Agence Spatiale Européenne,* 29 REVUE FRANÇAISE DE DROIT AÉRIEN ET SPATIAL 259 (1975); M. Bourély, *L'Agence Spatiale Européenne.* 1 ANN. AIR & SPACE L. 183-196 (1976); M. Bourély & G. Lafferranderie, *Les organisations de l'espace,* Jurisclasseur Droit International, Fasc 195; J. Chappez, *La création de l'Agence Spatiale Européenne,* ANNUAIRE FRANÇAIS DE DROIT INTERNATIONAL 801 (1975); H. Kaltenecker, *La nouvelle Agence Spatiale Européenne,* 17 PROC. COLLOQ. L. OUTER SPACE 126-137 (1974); H. Kaltenecker, *The European Space Agency in* 1 MANUAL ON SPACE LAW 259 (N. Jasentuliyana ed., Oceana 1979); R. F. von Preuschen, *The European Space Agency,,* 27 INT'L & COMP. L. Q. 46-60 (Jan. 1978).

12 On optional programmes: G. Lafferranderie, *Considérations sur la procédure d'engagement des programmes facultatifs,* ESA Bull. 53 (No.51, 1987); G. Lafferranderie, *Les Programmes Facultatifs;* Proceedings of the International Colloquium on "The Implementation of the ESA Convention - Lessons from the Past," held on October 23 and 26, 1993 in Florence, Italy (Nijhoff, 1994).

13 *See* art. XXII of the Liability Convention, art. VII of the Registration Convention, art. 16 of the Agreement Governing the Activities of States on the Moon and Other Celestial Bodies - adopted by the U.N. Gen. Assembly on December 5, 1979, *opened for signature* on Dec. 18, 1979, *entered into force* July 11, 1984 U.N. Doc. A/RES/34/68 (1979) [hereinafter Moon Agreement].

*Observer Status in UNCOPUOS*

The forerunner of ESA, ESRO, was granted an observer status in the United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS) in 1973.[14] Since then, ESA has participated actively in the discussions at the UNCOPUOS and functioned as a forum for coordinating the Member States positions via its International Relations Committee.

*ESA and the Space Treaties*

The Outer Space Treaty is only open to signature by States. However, Article 6 of this treaty provides that its provisions should be respected by international organizations. The four other space treaties concluded after the Outer Space Treaty also contain provisions for intergovernmental organizations. For example, the Rescue Agreement explicitly provides for an Intergovernmental Organization's being considered as a "launching authority." To obtain such recognition, it is required that the majority of the Member States of that organization be States Parties to the Rescue Agreement and to the Outer Space Treaty and that the intergovernmental organization declare its acceptance of the rights and obligations provided for in the Rescue Agreement.[15] A similar provision can be found in the Liability Convention.[16] Notably, States Parties to these Treaties have an obligation to ensure that the intergovernmental organization of which they are a Member will deposit such a declaration.[17] ESA[18] has submitted its declaration of acceptance of both treaties to the United Nations Secretary General and the Depository Governments.[19] The same procedure was followed with regard to the Registration Convention,[20] where Article VII provides for application of the provisions to intergovernmental organizations. The Moon Agreement has a similar provision in Article 16, but has not yet been ratified by the majority of ESA Member States.

For many years, ESA was the only international organization to have accepted the space treaties, thereby demonstrating its intent to

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14 *See* ESRO General Report of 1973. As an example, the Secretariats of ESRO and ELDO produced a legal analysis based on the "functional approach" on the definition/delimitation of outer space.

15 Art. 6 of the Rescue Agreement.

16 Art. XXII of the Liability Convention.

17 For example, in the Liability Convention, art. XXII, para. 2 provides that: "States members of any such organization which are States Parties to this Convention shall take all appropriate steps to ensure that the organization makes a declaration in accordance with the preceding paragraph."

18 More precisely, it was ESRO who declared the acceptance of both Conventions and ESA, in accordance with article XIX of the ESA Convention, took over these rights and responsibilities.

19 *See* M. Bourély, *L'Agence Spatiale Européenne et le Droit de l'Espace*, in ESA Bull. (13 May 1978).

20 *See* G. Lafferranderie, *L'application par l'Agence Spatiale Européenne, de la Convention sur l'immatriculation des objets lancés dans l'espace extra-atmosphérique*, 11 ANN. AIR &SPACE L. 229 (1986).

comply with the fundamental principles of space law. Further, not only did ESA declare its acceptance of the above space treaties, but it also makes sure that they are properly implemented. For example, in December 1977 ESA adopted internal rules, in the form of a Council Resolution, to implement the provisions of the Liability Convention. In line with these fundamental principles, one may also note ESA's practice of clearing orbits of dead satellites (OTS, Olympus).

*Participation in the Work of ITU*

ESA, through its Member States, is an active participant at the ITU World Administrative Radio Conferences (WARC)[21] and is involved in the discussions on allocations of frequencies and orbital positions due to its own need for frequencies for its various satellites and missions.[22] The most prominent example was the Agency's Olympus programme which provided, amongst others, experimental regional European DBS services for its participating Member States.[23] Recently, the Additional Plenipotentiary Conference held in Geneva in December 1993 adopted a proposal from the High Level Committee regarding the status of intergovernmental organizations operating satellite systems which can now also participate in Plenipotentiary Conferences. ESA requested to be recognized as an intergovernmental organization operating satellite systems.

*ESA and the Remote Sensing Principles*

Another area where ESA is involved in the UN discussions is that of the legal principles for remote sensing satellites. ESA operates the ERS-1 satellite which provides a permanent flow of information on the Earth's environment. In the ERS-1 Declaration[24] the participants explicitly stated, in advance of their approval, that they accept and apply the United Nations General Assembly Remote Sensing Principles.[25]

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21 For example, during the WARC-92 in Malaga-Torremolinos, ESA participated as an observer in the work of the Conférence Européenne des Postes at Télécommunications (CEPT) Ad Hoc Group WARC-92 which prepared European Common Provisions (ECP's) for the Conference.

22 In 1980, ESA established the Space Frequency Coordination Group (SFCG), a coordination group of the different space agencies for areas as space research, earth exploration, meteorological satellites and space operations. ESA status at the ITU is comparable with an observer status.

23 Bourély states that "On the basis of agreements with ESA, the European broadcasting entities using the satellite were responsible for observing the principles of Resolution 37/92 on international satellite television broadcasting." *See* M. Bourély, *Space Law and the European Space Agency, in* SPACE LAW DEVELOPMENT AND SCOPE 87 (N. Jasentuiliyana ed. Praeger 1992).

24 M. Ferrazzani, *The Legal Framework for the Use of ERS1 Data,* ESA Bull. 104 (No. 68, 1991).

25 *See* Annex C to the ERS Declaration. For the principles, *see* UNGA Res. 41/65 (Jan 27, 1987).

## ESA AND THE LAW OF SPACE ACTIVITIES

### *ESA Telecommunication Programmes* [26]

ESA's first experimental telecommunications satellite, the Orbital Test Satellite, was placed into orbit in May 1978 and provided pre-operational satellite communications capacity for the European PTT administrations. After the OTS, ESA developed the European Communication Satellite (ECS) series for the Eutelsat organization. ESA also developed the MAROTS satellites for the INMARSAT organization and launched the Olympus DBS satellite in 1989.[27] As a result of these telecommunication activities, ESA acquired a central position in Europe with respect to the development of the European satellite telecommunication industry.

ESA does not itself notify its communication satellites to apply for orbital positions and frequencies at the ITU. It is the French Government who fulfils this role on behalf of ESA. ESA studies the feasibility of certain frequencies and orbits for its future planned communication satellites. The Agency's telecommunications programme has recently been reorganized and is now being carried out under the name of Advanced Research in Telecommunications Systems (ARTES).[28] ESA has now started a discussion on the future of the European satellite telecommunications industry and the influence of the ongoing liberalization of this market. A working group of the Agency's Joint Communications Board (JCB) was established to examine the issue. The group's work was completed at the end of 1993 upon its submittal of its final report to the ESA Council.[29]

In line with its projects for non-geostationary-orbit satellites,[30] ESA has also started considering the legal implications of such projects.

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26 For an analysis of the legal issues at the beginning of the Telecommunication Programmes of ESA, *see* W. M. Thiebaut, *Legal Framework of Communications Programs in the European Space Agency,* 5 MICH. Y.B. INT'L STUD. 125 (1984); A. Pinglier, *L'ESA et l'évolution des télécommunications spatiales en Europe,* AÉRONAUTIQUE ET ASTRONAUTIQUE 57 (No. 154-155, 1992).

27 M. Bourély, *Olympus, le satellite européen de télévision directe,* 12 ANN. AIR & SPACE L. 193 (1987); P.H. Tuinder and O.M. Ribbelink, *ESA and the Development of Space Law, The Olympus Programme, in* 32 PROC. COLLOQ. L. OUTER SPACE 312 (1991).

28 The ARTES programme was approved by the ESA Council on March 25, 1993.

29 Full name of the Report is "Europe at the Cross-Roads: The Future of its Satellite Communications Industry," ESA SP-116, Nov.1993. The Report concludes that the R&D support given by ESA to European industry against the background of an increasing liberalization and international competition is inadequate and that satellite communications have to be seen as a fundamental economic pillar of the ESA programme.

30 For example, Archimedes an ESA project which plans to provide mobile communications and digital sound broadcasting from a satellite in a Highly Elliptical Orbit.

*Launching and Launching State*

ESA developed the ARIANE (ARIANE 1) launcher, which performed its first successful launch in December 1979, as well as the launch pad at Kourou, within the confines of the European Space Centre and associated tracking stations. The legal basis for the development of this launcher by ESA and the commercialisation by the private company, ARIANESPACE, is a good illustration of the way space law provisions influence ESA's activities[31] and of the particularly complex situation created by the interconnections between the ESA regime and the private regime under which the activities of Arianespace are carried out.

The execution of the development phase, which started in 1973 under an Arrangement concluded between several ESRO Member States, was carried out by the French Space Agency, CNES, under the direction of ESA. The commercialization of the ARIANE launcher was entrusted by the participating States to ARIANESPACE, a French private company created in 1979.[32] To allow for such commercialization, a Declaration containing the rights and obligations of the ARIANESPACE company was accepted by Member States.[33] ESA's relationship with ARIANESPACE is implemented through a Convention between Arianespace and ESA which was signed in May 1981, and renewed in 1990 following renewal of the Declaration of Certain Governments on the Ariane Production Phase. It is stipulated in this Convention that ESA has the right of access to all company documents as well as a consultative vote on the Board of ARIANESPACE.

In the Arianespace Declaration and the Convention between ESA and Arianespace, it is explicitly provided that the participants will respect the ESA Convention and that the activities of Arianespace will be carried out for peaceful purposes.[34] To control the sales of Arianespace and especially whether the use of the launcher will be for peaceful purposes, a Sales Control Committee has been created. All the States party to the Ariane Declaration are represented on this Committee. Should the Committee come to the conclusion that a planned sale of the launcher is not in accordance with the peaceful uses principle, the Committee can prohibit Arianespace from carrying out this sale. Such prohibition is executed by the French government, since France is the responsible State according to Article VI of the Outer Space Treaty.

The legal arrangements for Ariane launches illustrate well the influence of space law and contract law provisions. All Ariane launches take place at the Centre Spatial Guyana (CSG), which is French territory. The CSG also includes outside facilities. Within the perimeter of the CSG, ESA had constructed the Ariane Launching Pad, normally referred to as

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31 For an analysis of Arianespace and its relations with ESA, CNES and the French government, *see* V. Kayser, *Les Services Commerciaux de Lancement de Satellites - Aspects Juridiques -, in* La Lettre du C.E.R.D.I, Janvier 1993.

32 Shareholders of ARIANESPACE are the CNES (34.2%), European space industries (60.8%) and European banks (5%).

33 Declaration of Certain Member States Concerning the Production Phase of ARIANE, 14.1.80. Notably this is not a legal instrument (Declaration) in accordance with Article V.I-B of the ESA Convention, but a separate international agreement.

34 In accordance with art. II of the ESA Convention.

ELA.[35] The French government and ESA have concluded two types of agreements, one on the access and use by ESA of the CSG and its facilities and one on the functioning of the ELA. Two types of launching can be distinguished, namely, the qualification flights under the ESA research programme, and the operational, commercial launching by ARIANESPACE, with ESA as a paying customer. According to the provisions of the space treaties, there are in this particular situation several potential launching states: France, because the launch is performed on French territory and uses French facilities; ESA, because the launch pad was developed by ESA; and finally the client's, in the case of a commercial launch. These relations are governed by the CSG and ELA Agreements concluded between ESA and the French government as well as the Declaration on the Ariane Production Phase and the Convention between Arianespace and ESA. In the case of a commercial launch, relations are governed by contracts between Arianespace and its client, the insurance contracts, and the relationship described above between ESA, Arianespace and CNES.

*Earth Observation and Meteorology.*

In early 1972, ESA developed the experimental meteorological satellite Meteosat, in the absence of an adequate existing stucture able to carry out and finance such a programme.[36] Thereafter, ESA promoted the establishment of an organization independent of ESA to take over the operational activities. This gave birth in 1986 to a new International Organization EUMETSAT, to manage the operational meteorological satellite system developed by ESA. EUMETSAT, which has its headquarters in Darmstadt,[37] is now implementing preparations for the second generation of Meteosat satellites which will be developed by ESA and owned by EUMETSAT. The first meteorological programme of the Agency began in 1972 when ESRO decided to develop a pre-operational meteorological satellite system. The first Meteosat satellite was launched in 1977 and a protocol was adopted between Participating States to cover the first five years of exploitation by ESA.

Legal problems relating to the activities of EUMETSAT have mainly focused on the relationship between EUMETSAT and ESA[38] and how EUMETSAT can protect its meteorological data.[39] This last issue is related to the legal protection of satellite remote sensing data which will be discussed later in this article.

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35 ELA stands for Ensemble de Lancements Ariane.

36 G. Lafferranderie, *L'apport du programme Météosat opérationnel sur le plan juridique,* ESA Bull. 23 (No. 35, 1983).

37 S. Courteix, *Eumetsat, l'Europe de la météorologie par satellite,* 29 ANNUAIRE FRANÇAIS DE DROIT INTERNATIONAL 624 (1983).

38 For example the Météosat Second Generation Agreement concluded in February 1994 between ESA and Eumetsat contains a provision on the procurement of launchers for the three satellites to be developed.

39 *See generally* I.H.PH DIEDERIKS-VERSCHOOR, AN INTRODUCTION TO SPACE LAW 143-44 (Kluwer 1993) who summarizes the only known (lower) Court case dealing with protection of EUMETSAT data. *See also* the proceedings of the 1st Eumetsat Workshop on Legal Protection of Meteorological Data, March 13-14, 1989, Darmstadt, Germany, published by Eumetsat in 1991.

As stated above, at the time of adoption of the ERS-1 programme, the participating States explicitly declared their acceptance of the UNGA remote sensing principles. ESA's data policy reflects this through the adoption of two main principles: provision of the data on an open and non-discriminatory basis in accordance with the UNGA principles, and the claim of ownership of the ERS-1 data. This policy has been implemented in more than 20 agreements for the direct reception of ERS-1 data by third parties[40] and in several hundred distribution arrangements.

Currently, ESA is participating in the CEOS[41] to define data policy principles with other space agencies for the Global Earth Observation System. Furthermore, a special Earth Observation International Coordination Working Group (EO-ICWG) has been established by the four space station partners to coordinate their respective roles in the management of remote sensing data and definition of a data policy.

*Space Station*

The agreements creating the cooperation between ESA, Canada, Japan and the USA for the International Space Station can be divided into two phases: Phase B and the development phase. Phase B is covered only by MOU's, for example the MOU signed in 1985 at Agency level between NASA and ESA. The Intergovernmental Agreement on the Space Station (IGA)[42] between the States cooperating in this joint project, as well as the MOU's between NASA and each of the cooperating Agencies, which covered the development phase of this project, were signed in 1988. On the same day, an Arrangement was signed by the twelve states concerned (except Japan) to provisionally apply the IGA pending its approval or ratification.

The IGA and MOU's, which are now to be amended due to the potential participation of Russia as a new Partner, and the redesigned "Station Alpha," is the first general international implementation of principles of the space treaties and national laws of the Partners.[43] The IGA contains original clauses concerning cross-waiver, jurisdiction and control, exchange of information and data, criminal jurisdiction,[44] and intellectual property rights[45] on board the space station.[46] A code of

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40 *See* M. Ferrazzani, *ESA Data Policy,* ECSL News (9, June 1992).

41 Committee on Earth Observation Satellites (CEOS) established at the Economic Summit of the Group of Seven in Washington, 1984. ESA is a full member of CEOS.

42 Signed in September 1988. A. Farand, *La station spatiale et son régime juridique*, 15 ANN.AIR & AND SPACE L. 309 (1990); G. Lafferranderie, *Les Accords relatifs à la station spatiale internationale,* 22 REVUE GÉNÉRALE DE DROIT INTERNATIONAL PUBLIC 317 (No. 2, 1989).

43 As commonly known, one innovation in the IGA is the introduction of the "Partner"; USA, Canada, Japan and the European Partner representing the nine participating ESA Member States.

44 Criminal jurisdiction will be for the US when an act endangers the whole space station

45 A.Vahrenwald, *Intellectual Property on the Space Station Freedom,* 15 EUROPEAN INTELLECTUAL PROPERTY REV. 318 (No.9, 1993); A. M. Balsano, *Industrial Property Rights in Outer Space: The Space Station International Governmental Agreement (IGA) and the European Partner,* 35 PROC. COLLOQ. L. OUTER SPACE 216-

conduct for the astronauts working on board the Space Station is still to be drafted. In conjunction with the civil status of ESA and its Convention, one must note the exchange of correspondence between the heads of the negotiations on the civil character of the Space Station.

*Space Debris*

ESA addressed the problem caused by space debris for the first time in a publication that appeared in 1988.[47] The Council Working Group concluded that ESA had to take immediate action in order to avoid "irreversible developments or expensive consequences." Furthermore, it advised ESA to take the necessary organizational, technical, and institutional steps to counter this threat to space flight and to seek cooperation with other concerned parties.[48] In the same year, the Council of ESA adopted a Council Resolution with respect to the space debris issue.[49] The Council also adopted an activity programme with respect to space debris[50] and formulated the following objectives: to minimize the creation of space debris in order to ensure free access to space and reduce the risk for manned and unmanned space flight; to reduce the risk on ground due to reentry of space objects; to reduce the risk for geostationary satellites; to acquire data on space debris in order to assess the extent of the problem and its consequences; and to study the legal aspects of space debris.[51]

The ESA Space Debris Advisory Group (SDAG) was established in 1989. This group consists of experts from the ESA Member States and is, amongst other objectives, to provide a European Forum to exchange information on space debris matters. As space debris is largely an international problem, international coordination of efforts was considered to be important. An interagency committee was established in October 1993 and the participating members are ESA, Japan, NASA, and the Russian Space Agency.[52]

According to the reports published by the SDAG, the main problem is that there are no adequate provisions in the outer space treaties that deal with the prevention of pollution in outer space. Moreover, it was concluded that the provisions of the Registration Convention were insufficient for identifying the origin of some of the orbital debris. In 1993, ESA organised the first European conference on space debris in

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28 (1992).

46 *See* Article 21 IGA and the discussion later on Intellectual Property Rights (IPRs) in outer space.

47 Space Debris - The report of the ESA Space Debris Working Group, Doc. ESA SP-1109 (Nov. 1988),

48 *Id.* at 68.

49 Doc. ESA/C/LXXXVII/Res.3 (final, 29 June 1989) in ESA, BASIC TEXTS, vol. IV.

50 Doc. ESA/C(89)24, rev.1, ESA Activities for Space Debris.

51 *Id.*

52 Full name of the group is Inter-Agency Space Debris Coordination Committee (IADC) and its terms of reference were accepted at the second IADC Meeting in Kalingrad/Moscow on October 25-26, 1993.

Darmstadt, Germany.[53]

*Remote Sensing Intervention for European Parliament.*

As stated above, soon after the start-up of the ERS-1 programme, problems arose with respect to the legal title on whose basis protection could be assumed by the remote sensing satellite operator. For ESA, this problem became apparent when agreements had to be concluded with third parties for the reception of ERS-1 data. Although ownership of the data was claimed by ESA in these agreements, a clear legal title for this ownership was, and still is, lacking. This was the reason for the European Centre for Space Law's (ECSL) study, commenced in 1990, of the legal problems of protecting remote sensing data.[54] In 1991 a workshop was organized in Frascati where the results of this study were presented. The general conclusion was that the current legal framework is essentially inadequate and that a clear, precise, consistent and harmonized legal regime is necessary for the optimal exploitation of Earth observation data. Following this workshop, a new study continuing the work carried out in the ECSL study was issued by the European Commission at the end of 1991.[55] The purpose of this study was, amongst others, "to provide a comparative assessment of the main legal options possible and make recommendations regarding the steps towards a legal regime and the legislative measures that would best secure the interests of the providers and of the users of the data and the derived information."[56] This study, jointly managed by European Commission, ESA and ECSL representatives, was carried out in 1992 by a team of consultants under the direction of Professor Gaudrat of the University of Poitiers.

After reviewing the European legislations on protection of satellite data, the study confirmed that adequate protection is missing and that, although in some cases and in some European countries protection can be provided, the policy differs from country to country. The solution proposed in the study is to place the protection of satellite data under the umbrella of the proposed (European Commission) Council Directive on the legal protection of databases.[57] The attraction of this draft directive lies in the fact that it introduces a *sui generis* protection of (electronic) databases, so that protection is no longer dependent on creativity or authorship as under the existing scheme of protection by copyright or author's right.[58] However, to make this draft directive applicable to remote sensing data, the definition of a database[59] and the definition of

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53 *See* proceedings of the first Euroepan Conference on Space Debris, Darmstadt, Germany, 5-7 April 1993, Doc. ESA SD-01 (1993).

54 The study was carried out by J. D. Dupuy, a Toulouse based consultant and former legal adviser of SPOT.

55 Full title of this study: Conditions of Access to Earth Observation Data: Legal Aspects, Dec. 1992.

56 *Id.* (Terms of reference of the study).

57 Doc. COM(92) 24 final-SYN 393 (13 May 1992).

58 At the same time this draft directive will create a harmonized regime in all European Union Member States and EFTA countries.

59 Art. 1, subpara. 1 of the draft directive defines a database as "a collection

"unfair extraction" need to be clarified.[60] Following this analysis, proposals were made by the ESA Executive, after having consulted Prof. Gaudrat, to change the draft Directive accordingly.[61] ESA was invited to give its view on the draft Directive at a hearing of the Legal Committee of the European Parliament on 17 March 1993.[62] The Legal Adviser of ESA presented the Agency's view on the draft Directive, its importance for European remote sensing activities, and the need to alter the definitions in order to make it applicable to remote sensing data.[63]

The draft Directive is still being discussed by the Council of the European Union, where the members are trying to obtain a common position which would enable the Council to send a consolidated text to the European Parliament for approval.[64]

*Intellectual Property Rights (IPR's) in Outer Space*

Due to the planned implementation of the International Space Station, ECSL's attention was drawn to the problem of the protection and granting of patents in outer space. This resulted in a research project of ECSL to find out whether European industry and policy makers indeed considered Intellectual Property Rights in outer space a subject of concern and worth examination.[65] The results of this research were presented at a 1993 workshop in Madrid where some sixty specialists (including a representative from the World Intellectual Property Organisation) participated. Key issues were the difference between the US (first to file) and the European (first to invent) systems, the lack of specific European legislation making patent law applicable to outer space activities,[66] the influence of the micro-gravity environment on the criterion of novelty, inventive step and industrial application, and the infringement of a patent

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of works or materials arranged, stored and accessed by electronic means, and the electronic materials necessary for the operation of the database such as its thesaurus, index or system for obtaining or presenting information; it shall not apply to any computer program used in the making of the operation of the database."

60 Art. 1, subpara. 2 of the draft directive defines "the right to prevent unfair extraction" as the right of the maker of a database to prevent acts of extraction and re-utilization of material from that database for commercial purposes.

61 A letter was sent by the Director of Administration of ESA on December 4, 1992, to the Commission, explaining the interest of ESA and its Member States to have a harmonized European regulation protecting remote sensing data.

62 This was the first time ESA was invited to present its views at a European Parliament (Committee) Meeting.

63 Full text of the intervention in Doc. PE 204.234/4.

64 At the time of writing a modified version of the draft directive has been issued by the European Commission which appears to go into the direction ESA was proposing. For example the word "works" in the definition of a database has been deleted.

65 For more details, *see Proceedings of the first ECSL/Spanish Centre for Space Law Workshop on Intellectual Property Rights in Outer Space, Madrid, May 26, 1993* ( ECSL publ. 1993).

66 For the U.S., see the Patents in Outer Space Act.

in outer space.[67] The conclusion drawn from the discussions was that Europe's participation in the International Space Station programme will give rise to specific problems in view of the (ten) competing national patent legislations applicable to inventions realized on the European module. The question of how European participants should ratify the IGA so that national patent law can be applicable[68] was also discussed.

In general, the workshop concluded that there was a need for worldwide harmonization of patent laws applicable to outer space activities.[69] This harmonization should take into account legislation applicable to other (non-space) activities. The harmonization should be based on common rules to be found in existing national patent laws and should firstly take the form of an international code of conduct for micro-gravity activities.

Since worldwide harmonization was considered to be necessary, WIPO was asked to take the next step to identify ways of implementing these recommendations. This year a follow-up of the study is planned to be undertaken by ECSL/ESA, possibly in cooperation with WIPO.

*Astronauts Policy* [70]

With the International Space Station, the MIR space station, and the Shuttle flights, manned space flight is becoming more and more prominent in space activities. ESA is training its own team of astronauts[71] to fly on the space shuttle and on the Russian Soyuz capsules.

It is interesting to note that provisions in the space treaties constitute only a general regime applicable to astronauts, particularly with regard to situations of distress.[72] Some writers stated in this respect that astronauts, as envoys of mankind in such exceptional circumstances, must enjoy special treatment, and that astronauts will often be part of an international crew and work in international territory. As a result, questions have to be solved with regard to which (national) legislation is applicable, what is the responsibility of the state who has jurisdiction and control, *etc.* There is also the problem of the various terms used to describe astronauts, namely, "personnel,"[73] "envoy of mankind," and "representative."[74] Whatever definitional problems have arisen in the legal doctrine, multinational space crews have constituted a certain

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67 *Op. cit. supra* note 39, at 74.

68 As done by Germany which considers for the applicability of its patent law that the work on the European module of the space station will be considered as having taken place on its own territory

69 Here it was considered that the IGA only deals with the International Space Station and thus is limited in scope

70 For more details, *see* G. Lafferranderie, *Pour Une Charte de L'Astronaute,* 12 ANN. AIR & SPACE L. 263 (1987).

71 At the European Astronaut Centre (EAC) in Cologne and in the USA and Russia.

72 *See* the Rescue Convention.

73 Art. VIII of the Outer Space Treaty.

74 *See also* M. Bourély, *Towards a Convention on the Legal Status of Manned International Space Flights,* 22 PROC. COLLOQ. L. OUTER SPACE 59 (1980).

practice and states or agencies conclude special agreements to arrange for the participation of other nationals in their space transportation and station missions.[75] In general, however, the regime can be described as *ad hoc* and not sophisticated enough for the time when astronauts will become regular visitors of space stations or passengers of aerospace planes (ASP) or other new single stage to orbit systems.

An attempt has been made to draft new legislation expressly for manned spaceflight.[76] Some writers go even further and conclude that there is a need for a "second generation space law" which would deal in much more practical terms with code of conduct for astronauts who could be journalists or maybe even tourists in the future. It goes without saying that in these cases they can hardly be considered as the envoys of mankind envisaged during the drafting of the space treaties.[77]

In conjunction with the Spacelab programme, and the Hermes programme, ESA has studied the definition of space objects, as well as the legal implications of choices made.[78]

*ECSL Activities* [79]

Many activities promoting the development of space law are conducted by the European Centre for Space Law (ECSL), an independent research centre operating under the chairmanship of the legal advisor of ESA. The Centre has one staff member (an ESA young graduate) and is supported mainly through funding from the Agency. Originally, the Centre was to act as a forum for exchanges of views by its European members. However, the Centre developed into a kind of small European "think-tank" on space law matters. The Centre publishes reports and a newsletter and organises workshops, research, fora for discussions and a Summer Course[80] on space law and policy which is supported by the European Union ERASMUS programme. Moreover, the Centre manages a unique database containing documents in electronic form on space treaties, ESA basic documents and space law policy papers which are accessible to its members. The Centre has currently over four hundred individual members from all interest groups of the ESA Member States.[81] ECSL also reports to the United Nations on its educational activities in the framework of the UN

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75 *See* the ESA/NASA MOU for Space Lab and the flight of Nicholier as an ESA astronaut for the Mission to repair the Hubble telescope.

76 Draft Convention on Manned Space Flights, a joint project of Prof. Böckstiegel, Prof. Vereschetin and Prof. Gorove. For the text, *see* MANNED SPACE FLIGHT (K.-H. Böckstiegel ed. Cologne 1993).

77 *See* Y. Hashimoto, *The Status of Astronauts: Toward the Second Generation Space Law,* 36 PROC. COLLOQ L. OUTER SPACE (in print).

78 Proceedings of the Colloquium *"L'avion spatial et le droit,"* REVUE FRANÇAISE DE DROIT AÉRIEN ET SPATIAL 427-571 (No. 4, 1991).

79 See for more details the ECSL Newsletters and the two Biennial Reports of 1991 and 1993.

80 *The Second ECSL Summer Course on Space Law and Policy, Toulouse, September 6-18, 1993,* 21 J. SPACE L. 170 (1993).

81 See the two ECSL Biennial Reports describing its activities in the periods 1989-1991 and 1991-1993, both available from the ECSL Secretariat.

Decade of International Law.[82] Upon the initiative of ECSL, National Points of Contacts (NPOC's) have been established in Belgium, France, Germany, Italy, Spain, Sweden, the Netherlands and the United Kingdom.

*Trend Towards Multilateralism and MOU's*

Since 1992, there has been an emphasis within ESA on international cooperation for the execution of space programmes.[83] Apart from the political changes in Europe, the main motive for emphasizing international cooperation is the fact that space programmes are becoming more complex and costly. Cooperation allows not only the burden of cost to be shared, but also the expertise and facilities of the different partners. Cooperation is normally made possible through MOU's and agreements (at Agency level).

*Influence of Charters and Development of "Soft Law"*

Traditionally, cooperation between ESA and other organizations or institutions is concluded through formal agreements adopted by Council, or through the granting to certain States of the status of cooperating State, or associated State. Progressively, and in keeping with the general trend in international law, ESA is turning to soft law and is entering into cooperation where legal terms are defined in Charters or exchange of letters. Such Charters or texts set up the bodies adapted to the conduct of the programmes concerned and provide for the modalities under which these programmes will be executed and the information exchanged. As an example of such mechanisms, one may refer to the Inter-Agency Coordination of Solar-Terrestrial Science Projects,[84] where a Consultative Group was established which coordinated the six space missions to Halley's Comet undertaken by its four Member Agencies (Intercosmos, ISAS, NASA and ESA).

*Coordination with other European Organizations.*

ESA was the result of the European Space Conferences in the 1970's when European States came to the conclusion that the efforts of the CETS, ELDO and ESRO had to be merged in order to establish a coherent, balanced space research and development agency which, apart from science activities, also should be mandated to carry out application activities. Since the establishment of ESA, EUTELSAT, EUMETSAT and ARIANESPACE have been created as a kind of "spin-off" from ESA activities, proving that the initial work of ESA (in applications) in the area of telecommunications, meteorology and launchers was so advanced that operational organizations

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82 *See* UNGA Res. 45/40 of 20. Nov. 1989.

83 *See* ESA/C-M/CIV/Res.2 (Final) Resolution on International Cooperation (adopted on 10 November 1992 at the Council meeting at ministerial Level in Granada).

84 R. Reinhard, *The role of the IACG and its Associated Working Groups,* ESA Bull. 81 (No. 46, 1986); R. Reinhard, I ESA Bull. 8 (No.51, 1987) (terms of reference of the IACG are also reproduced in annex to this article).

could carry out these activities.

However, this well balanced situation, where ESA was the focal point of European space research and development activities, and at the same time building up the space industry in its Member States through its geographical return policy, is now changing. The European Union made its entry into the domain of space policy and legislation with its policy of liberalization of satellite telecommunications,[85] affecting the position of the telecommunications industry and EUTELSAT, and with its Single European Act (SEA) in 1987 which established a mandate in the area of Research and Development.[86] Apart from the enlarged mandate of the European Union (including a common defense and security policy), political and economic changes also produced this new situation for ESA.

Exploratory contacts have been established between the European Commission and ESA to seek a basis for a framework agreement between the two international organizations which would provide for improved coordination in regards to the respective research and development programmes and policy for space related activities.[87] Furthermore, ESA is asked to provide technical assistance to the Western European Union for the development of a European space-based observation programme.[88]

ESA, as an international cooperative for space research and development activities, plays an important role in the further implementation of space law principles. Through its Member States, it is involved in UN discussions at COPUOS and ITU. As an actor it is confronted with the need for a practical translation of space law principles into clear legal provisions. Consequently, ESA is increasingly cooperating with the national space agencies and other international organisations to create regulations which will facilitate the activities of the Agency, its Member States, and the (European) industries and users of space services.

It is clear that international organizations, like ESA, can no longer be excluded from the process of creating and implementing space law principles. Indeed, were the space treaties to have been written today

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85 See for an overview of the questions related to the liberalization of the Satellite Telecommunication Market; The Implications of the CEC Green Paper on Satellite Communications in Europe, ECSL Proceedings of the ECSL/Dutch NPOC Workshop held at ESTEC on 27 Sept. 91

86 Due to the ratification of the Maastricht Treaty on the European Union, The European Parliament is becoming actively involved in the Research and Development policy as it will have the right of co-decision (before the Maastricht Treaty it had only advisory power), *see* the procedure of Article 189(b) *as amended* by the Treaty on the European Union

87 On relations EC/ESA *see*: K. Madders, W. Thiebaut, *Two Europes in One Space: The Evolution of Relations Between ESA and the European Community in Space Affairs*, 20 J. SPACE L 117 (1992); P. H. Tuinder, *The Implications of the Emerging European Community's Regulatory Policy Role in the Field of Space Activities*, 18 ANN AIR & SPACE L. 331 (Pt. I, 1993); D. L. Boquet, *Droit Communautaire et Espace*, Jurisclasseur Droit Communautaire, fasc. 1240 (Nov. 1993).

88 See the Western European Union Recommendation on the Development of a European space-based observation system which was adopted by its Assembly in December 1993. It is recommended that "the Council will contact the European Space Agency without delay in order to promote close cooperation between the two organizations in space matters" (Recommendation 5).

rather than three decades ago, they would have to take account of international organizations, such as ESA, and would be worded differently. The space treaties are a typical product of their time, but in the present climate additional provisions should be worked out concerning such concepts as registration, liability, jurisdiction and control. It is also important to underline the role of international organizations such as ESA in the implementation of basic space law principles, as well as complementing space law provisions with what we would like to call soft law.